

# Eugene Katsevich

Assistant Professor of Statistics and Data Science

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Academic Research Building 311  
265 South 37th Street  
Philadelphia, PA 19104-1686

Phone: (650) 250 3331  
Email: [ekatsevi@wharton.upenn.edu](mailto:ekatsevi@wharton.upenn.edu)  
Homepage: <https://ekatsevi.github.io/>

## Employment

**University of Pennsylvania**, Department of Statistics and Data Science, Assistant Professor, 2020-present.

- Member of the graduate group in Applied Mathematics and Computational Science (AMCS)
- Member of the graduate group in Genomics and Computational Biology (GCB)

**Carnegie Mellon University**, Department of Statistics and Data Science, Postdoctoral Researcher, 2019-2020.

Advisors: Kathryn Roeder and Aaditya Ramdas.

## Education

**Stanford University**, Ph.D. in Statistics, 2019.

Thesis Advisor: Chiara Sabatti.

**Princeton University**, A.B. in Mathematics (with Highest Honors), 2014.

Thesis Advisor: Amit Singer.

## Publications

### *Working manuscripts*

27. A. Chakraborty, J. Lee, **E. Katsevich**. Comparing three learn-then-test paradigms in a multivariate normal means problem. Available on [arXiv](#).
26. T. Barry, Z. Niu, **E. Katsevich**, X. Lin. The permuted score test for robust differential expression analysis. Available on [arXiv](#).
25. T. Barry, J. Deutsch, and **E. Katsevich**. Hands-on single-cell CRISPR screen analysis. e-book available [online](#).

### *Manuscripts under review*

23. Z. Niu, J. Ray Choudhury, and **E. Katsevich**. The conditional saddlepoint approximation for fast and accurate large-scale hypothesis testing. Under review at *Biometrika*. Available on [arXiv](#).
22. A. Chakraborty, J. Zhang, and **E. Katsevich**. Doubly robust and computationally efficient high-dimensional variable selection. Under review at *JRSS-B*. Available on [arXiv](#).

21. J. Ray, E. Jagoda, M. U. Sheth, J. Galante, D. Amgalan, A. R. Gschwind, C. J. Munger, J. Huang, G. Munson, M. Murphy, E. Mattei, T. Barry, V. Singh, A. Baskaran, H. Kang, **E. Katsevich**, L. M. Steinmetz, and J. Engreitz. An unbiased survey of distal element-gene regulatory interactions with direct-capture targeted Perturb-seq. Under review at *Nature Genetics*. Available on [bioRxiv](#).
20. P. J. Law, J. Smith, J. Vijayakrishnan, T. Barry, D. Chubb, B. Kinnersley, M. Mandelia, A. J. Cornish, **E. Katsevich**, and R. S. Houlston. Identification of enhancer-gene regulatory interactions in colorectal cancer identified through genome-wide CRISPRi perturbations. Under review at *Genome Biology*.

### *Published or forthcoming papers*

20. L. Deutsch and **E. Katsevich**. Location tests with noisy proxies for latent variables. *Statistics and Probability Letters*, 2025. Available on [Science Direct](#).
19. M. Conery, J. A. Pippin, Y. Wagley, K. Trang, M. C. Pahl, D. A. Villani, L. J. Favazzo, C. L. Ackert-Bicknell, M. J. Zuscik, **E. Katsevich**, A. D. Wells, B. S. Zemel, B. F. Voight, K. D. Hankenson, A. Chesi, and S. F. A. Grant. GWAS-informed data integration and non-coding CRISPRi screen illuminate genetic etiology of bone mineral density. *Genome Biology*, 2025. Available at [Genome Biology](#).
18. R. E. Yan, A. Corman, Z. Z. Gajic, R. Sam, M. Farid, I. Raimondi, S. Ganesan, **E. Katsevich**, N. Dahmane, J. P. Greenfield, and N. E. Sanjana. Pooled CRISPR screens with joint single-cell chromatin accessibility and transcriptome profiling. *Nature Biotechnology*, 2024. Available at [Nature Biotechnology](#).
17. T. Barry, K. Roeder, and **E. Katsevich**. Robust differential expression testing for single-cell CRISPR screens. *Genome Biology*, 2024. Available at [Genome Biology](#).
16. T. Barry, K. Roeder, and **E. Katsevich**. Exponential family measurement error models for single-cell CRISPR screens. *Biostatistics*, 2024. Available at [Oxford University Press](#).
15. Z. Niu, A. Chakraborty, O. Dukes, and **E. Katsevich**. Reconciling model-X and doubly robust approaches to conditional independence testing. *Annals of Statistics*, 52(3):895–921, 2024. Available at [Project Euclid](#).
14. J. A. Morris, Z. Daniloski, J. Domingo, T. Barry, M. Ziosi, D. A. Glinos, S. Hao, E. Mimitou, P. Smibert, K. Roeder, **E. Katsevich**, T. Lappalainen, and N. E. Sanjana. Discovery of target genes and pathways of blood trait loci using pooled CRISPR screens and single cell RNA sequencing. *Science*, 2023. Available at [Science](#).
13. J. Tian, X. Chen, **E. Katsevich**, J. Goeman, and A. Ramdas. Large-scale simultaneous inference under dependence. *Scandinavian Journal of Statistics*, 50(2): 750–796, 2023. Available at [Wiley](#).
12. **E. Katsevich**, C. Sabatti, and M. Bogomolov. Filtering the rejection set while preserving false discovery rate control. *Journal of the American Statistical Association*, 118(541): 165–176, 2023. Available at [Taylor and Francis Online](#).
11. **E. Katsevich** and A. Ramdas. On the power of conditional independence testing under model-X. *Electronic Journal of Statistics*, 16(2): 6348–6394, 2022. Available on [Project Euclid](#).
10. M. Liu, **E. Katsevich**, L. Janson, and A. Ramdas. Fast and powerful conditional randomization testing via distillation. *Biometrika*, 109(2):277–293, 2022. Available at [Oxford University Press](#).
9. T. Barry, X. Wang, J. A. Morris, K. Roeder, and **E. Katsevich**. SCEPTRE improves calibration and sensitivity in single-cell CRISPR screen analysis. *Genome Biology*, 22:344, 2021. Available at [Genome Biology](#).

8. **E. Katsevich** and A. Ramdas. Simultaneous high-probability bounds on the false discovery proportion in structured, regression, and online settings. *Annals of Statistics*, 48(6):3465–3487, 2020. Available at [Project Euclid](#).
7. M. Sesia, **E. Katsevich**, S. Bates, E. Candès, C. Sabatti. Multi-resolution localization of causal variants across the genome. *Nature Communications*, 11:1093, 2020. Available at [nature.com](#).
6. J. Zhu, Q. Zhao, **E. Katsevich**, C. Sabatti. Exploratory Gene Ontology Analysis with Interactive Visualization. *Nature Scientific Reports*, 9:7793, 2019. Available at [nature.com](#).
5. **E. Katsevich** and C. Sabatti. Multilayer Knockoff Filter: Controlled variable selection at multiple resolutions. *Annals of Applied Statistics*, 13(1):1–33, 2019. Available at [Project Euclid](#).
4. J. Anden, **E. Katsevich**, and A. Singer. Covariance estimation using conjugate gradient for 3D classification in cryo-EM. In *IEEE Int Symp Biomed Imaging*, New York, New York, April 2015. Available at [PubMed](#).
3. **E. Katsevich**, A. Katsevich, A. Singer. Covariance matrix estimation for the cryo-EM heterogeneity problem. *SIAM Journal on Imaging Sciences*, 8(1):126–185, 2015. Available at [PubMed](#).
2. B. Shi, **E. Katsevich**, B. Chiang, A. Katsevich, and A. Zamyatin. Image registration for motion estimation in cardiac CT. In *SPIE Medical Imaging*, San Diego, California, February 2014. Available at [SPIE digital library](#).
1. **E. Katsevich**, A. Katsevich, and G. Wang. Stability of the interior problem for polynomial region of interest. *Inverse Problems*, 28(6), 2012. Available at [PubMed](#).

## Software

- **sceptre**: R package for rigorously identifying regulatory relationships based on single-cell CRISPR screen experiments ([webpage](#), [manual](#)). This package is the first to be [recommended by 10x Genomics](#) for single-cell CRISPR screen analysis.
- **PerturbPlan**: Shiny app and companion R package for Perturb-seq and TAP-seq experimental design. It helps balance power and cost, with an emphasis on speed, flexibility, and interactivity ([web app](#), [R package](#)).
- **ondisc**: R that facilitates analysis of large-scale single-cell data out-of-core ([webpage](#)).
- **simulatr**: R package and Nextflow pipeline enabling portable and scalable numerical simulations ([webpage](#)).

## Recognition

- Claude Merion Faculty Scholar (2025-2026)
- Wharton Teaching Excellence award (2021)
- Jerome H. Friedman Applied Statistics Dissertation Award (2019).
- Statistics Department Teaching Assistant Award (2016).
- Hertz Fellowship (2014-2019).
- National Defense Science and Engineering Fellowship, declined (2014-2017).
- George B. Covington Thesis Prize in Mathematics (2014).

- Early election to Phi Beta Kappa (2013).
- Barry Goldwater Scholarship (2012).
- Shapiro Prize for Academic Excellence (2011, 2012).
- Freshman First Honor Prize (2011).

## Awards received by trainees

- Ziang Niu received the Lawrence D. Brown best paper award from the Wharton Statistics and Data Science department for “Reconciling model-X and doubly robust approaches to conditional independence testing.” (2024)
- Timothy Barry received Mihaela Serban Memorial Award for best poster from the Pittsburgh chapter of the American Statistical Association for “Robust differential expression testing for single-cell CRISPR screens at low multiplicity of infection.” (2023)
- Timothy Barry received ASHG Reviewers’ Choice Award for abstract on “SCEPTRE improves calibration and sensitivity in single-cell CRISPR screen analysis.” (2021)

## Teaching

### *Courses Taught at University of Pennsylvania*

- STAT 9610: Statistical Methodology (PhD level), Fall ’21, Fall ’22, Fall ’23, Fall ’24.
- STAT 4710: Modern Data Mining (Undergraduate level), Spring ’21, Fall ’21, Fall ’22, Fall ’23.

### *Courses Taught at Stanford*

- STATS 302: Qualifying Exams Workshop (PhD Level), Summer ’17.

## Training

### *Doctoral Student Supervision*

- Ziang Niu; jointly with Bhaswar Bhattacharya (Wharton Statistics and Data Science; Fall 2023 – present)
- Louis Deutsch (Wharton Statistics and Data Science; Summer 2023 – present)
- Timothy Barry; jointly with Kathryn Roeder (Carnegie Mellon University; Summer 2020 – Fall 2023)

### *Dissertation Committee Membership*

- Kaishu Mason (Wharton Statistics and Data Science; Spring 2024 – present)
- Abhinav Chakraborty (Wharton Statistics and Data Science; Spring 2024 – present)
- Jeffrey Zhang (Wharton Statistics and Data Science; Spring 2024 – present)
- Ganesh Chandrasekaran (UPenn Biostatistics; Fall 2023 – present)
- Shannon Laub (UPenn Bioengineering; Fall 2023 – present)

*Doctoral Student Research Mentorship (Informal)*

- Sophia Lu (Stanford University Statistics; Summer 2021)
- Abhinav Chakraborty (Wharton Statistics and Data Science; Fall 2021 – present)
- Jeffrey Zhang (Wharton Statistics and Data Science; Spring 2023 – present)
- Kaishu Mason (Wharton Statistics and Data Science; Spring 2023)
- Junu Lee (Wharton Statistics and Data Science; Fall 2024 – present)
- Yihui He (Wharton Statistics and Data Science; Fall 2024 – present)

*Master's Student Research Mentorship (Informal)*

- Jyotishka Ray Choudhury (Indian Statistical Institute; Summer 2023 – present)
- Ziang Niu (University of Pennsylvania AMCS; Spring 2022 – Summer 2023)

*Undergraduate Student Research Mentorship (Informal)*

- Winnie Dong (University of Pennsylvania; Spring 2024)

**Professional Service Activities**

- *Member*, NSF Review Panel (2024)
- *Organizer*, topic-contributed session on “Statistical analysis for pooled single-cell CRISPR screen data” at STATGEN conference in Pittsburgh (2024)
- *Member*, Statistics departmental postdoc selection committee (2021-22, 2022-23)
- *Member*, Statistics PhD admissions committee (2021-22, 2022-23, 2023-24)
- *Organizer*, Wharton statistics departmental seminar (Spring 2022, Spring 2024)
- *Reviewer* (39 manuscripts): Annals of Statistics (6), Journal of the Royal Statistical Society (Series B) (4), Journal of the American Statistical Association (5), Biometrika (3), Annals of Applied Statistics (1), Statistical Science (1), Proceedings of the National Academy of Sciences (2), Biometrics (2), Statistics and Probability Letters (1), Statistics and Computing (1), Journal of Computational and Graphical Statistics (2), Annals of the Institute of Statistical Mathematics (1), Journal of Multivariate Analysis (1), Electronic Journal of Statistics (1), Science (1), Nature Methods (2), Cell Systems (1), American Journal of Human Genetics (1), Genetics (1), Genome Medicine (1), BMC Bioinformatics (1).
- *Organizer*, Stanford statistics department orientation program for PhD students (2018).
- *Organizer*, Hertz West Coast Retreat (2017).
- *Academic chair*, Princeton Math Club (2012).
- *Head problem writer*, Princeton University Math Competition (2011).

## Presentations

### Invited talks

- *spaCRT: a resampling-free conditional randomization test.*  
International Conference on Statistics and Data Science in Nice, France, Dec. 16-19, 2024.
- *Scaling SCEPTRE to large single-cell CRISPR screens.*  
STATGEN 2024 in Pittsburgh, Pennsylvania, May 1-3, 2024.
- *Reconciling model-X and doubly robust approaches to conditional independence testing.*  
International Conference on Statistics and Data Science in Lisbon, Portugal, Dec. 18-21, 2023.
- *Robust differential expression testing for single-cell CRISPR screens.*  
EMBL-EBI industry workshop: Perturb-Seq data analysis and integration in Cambridge, MA, Jun. 21-22, 2023.
- *Robust differential expression testing for single-cell CRISPR screens.*  
New England Statistics Symposium in Boston, MA, Jun. 6, 2023.
- *Robust differential expression testing for single-cell CRISPR screens.*  
Impact of Genomic Variation on Function (IGVF) Consortium CRISPR focus group, May 19, 2023, held virtually.
- *Model-X versus doubly robust conditional independence testing.*  
Statistics Seminar at Stanford University, May 2, 2023.
- *Model-X versus doubly robust conditional independence testing.*  
Causal Inference Reading Group at Cambridge University, Apr. 28, 2023, held virtually.
- *Reconciling model-X and doubly robust approaches to conditional independence testing.*  
International Seminar on Selective Inference, Mar. 1, 2023, held virtually.
- *Statistical Analysis of Single Cell CRISPR Screens.*  
Joint Statistical Meetings in Washington, D.C., Aug. 8, 2022.
- *Statistical Analysis of Single Cell CRISPR Screens.*  
Statistics seminar at the Data Science and Operations department at the University of Southern California, Apr. 22, 2022, held virtually.
- *Statistical Analysis of Single Cell CRISPR Screens.*  
Biostatistics seminar at Yale University,  
Feb. 2, 2021, held virtually.
- *Statistical Analysis of Single Cell CRISPR Screens.*  
Open Insights in Biomedical Data Science Seminar at the University of Pennsylvania,  
Nov. 13, 2020, held virtually.
- *Finite-sample optimality and large-sample power analysis of the conditional randomization test.*  
Young Data Science Researcher Seminar at ETH Zurich, Nov. 6, 2020, held virtually.
- *Statistical analysis of single cell CRISPR screens.*  
Stanford Biostatistics Workshop, Oct. 8, 2020, held virtually.
- *The conditional randomization test in theory and in practice.*  
International Seminar on Selective Inference, Aug. 20, 2020, held virtually.
- *A theoretical treatment of conditional independence testing under Model-X.*  
Joint Statistical Meetings, Aug. 2-6, 2020, held virtually.

- *Multiple testing for modern data: structure, curation, and replicability.*  
U.C. San Diego Halicioglu Data Science Institute Special Seminar, Feb. 12, 2019.
- *Multiple testing for modern data: structure, curation, and replicability.*  
Rutgers University Statistics Seminar, Feb. 5, 2019.
- *Multiple testing for modern data: structure, curation, and replicability.*  
Wharton Statistics Seminar, University of Pennsylvania, Jan. 30, 2019.
- *Multiple testing for modern data: structure, curation, and replicability.*  
University of Chicago Statistics Colloquium, Jan. 24, 2019.
- *Controlling FDR while highlighting distinct discoveries, with applications to GO enrichment analysis.*  
Stanford University Biostatistics Workshop, Oct. 11, 2018.
- *Controlling FDR while highlighting distinct discoveries, with applications to GO enrichment analysis.*  
U.C. Berkeley Statistics and Genomics Seminar, Sep. 27, 2018.

### Invited discussions

- *"False discovery proportion envelopes with  $m$ -consistency"* by Iqraa Meah.  
International Seminar on Selective Inference, Nov. 25, 2024, held virtually.
- *"Simultaneous FDP bounds via knockoffs and closed testing"* by Jinzhou Li.  
International Seminar on Selective Inference, Jan. 25, 2023, held virtually.
- *"Detecting Multiple Replicating Signals using Adaptive Filtering Procedures"* by Jingshu Wang.  
International Seminar on Selective Inference, May 28, 2020, held virtually.

### Contributed talks

- *PerturbPlan: A web application for Perturb-seq experimental design.*  
Impact of Genomic Variation on Function Consortium Annual Meeting, Oct. 21-32, 2025 in Durham, North Carolina. One of 12 abstracts selected for podium presentation to entire consortium.
- *spaCRT: a resampling-free conditional randomization test.*  
Bernoulli-IMS World Congress in Probability and Statistics, Aug. 12-16, 2024 in Bochum, Germany.
- *A theoretical treatment of conditional independence testing under Model-X.*  
Bernoulli-IMS One World Symposium, Aug. 24-28, 2020, held virtually.
- *Simultaneous High-Probability Bounds on the False Discovery Proportion for Nested Sequences of Rejection Sets.*  
International Conference on Multiple Comparison Procedures, Dec. 13-15, 2019, in Taipei, Taiwan.
- *Gene Ontology enrichment testing: Reconciling FDR control with filtering.*  
Joint Statistical Meetings, Jul. 28-Aug. 2, 2018, in Vancouver, Canada.
- *The multilayer knockoff filter: Controlled multi-resolution variable selection.*  
International Conference on Multiple Comparison Procedures, Jun. 20-23, 2017, in Riverside, California.

*Poster Presentations*

- *Perturbation-expression association analysis in low-MOI single-cell CRISPR screens with SCEPTRE.*  
Biology of Genomes, May 9 - 13, 2023, in Cold Spring Harbor Laboratory, New York.
- *SCEPTRE improves calibration in single-cell CRISPR screen analysis.*  
Biology of Genomes, May 10 - 14, 2022, in Cold Spring Harbor Laboratory, New York.
- *Conditional resampling improves sensitivity and specificity of genome-wide CRISPR regulatory screens.*  
American Society for Human Genetics, Oct. 27–30, 2020, held virtually.
- *Controlling FDR While Highlighting Selected Discoveries.*  
International Conference on Multiple Comparison Procedures, Dec. 13–15, 2019, in Taipei, Taiwan.
- *Multi-resolution association analysis for exome-wide sequencing.*  
American Society for Human Genetics, Oct. 16–20, 2018, in San Diego, California.
- *Controlling FDR while highlighting distinct discoveries.*  
Workshop on Higher-Order Asymptotics and Post-Selection Inference, Sep. 8–10 2018, in St. Louis, Missouri.
- *Multilayer FDR control for genetic association studies.*  
Graybill Conference on Statistical Genomics and Genetics, Jun. 5–7 2017, in Fort Collins, Colorado.  
Best student poster award.
- *The multilayer knockoff filter: Multilayer FDR control for association studies.*  
Probabilistic Modeling in Genomics, Sep. 12–14, 2016, in Oxford, United Kingdom.



## Grant activity

### *Current*

| <b>Name</b>                                         | <b>Funding agency</b>       | <b>Period</b>          | <b>Type</b> | <b>Role</b> | <b>Annual direct cost</b> | <b>Annual indirect cost</b> |
|-----------------------------------------------------|-----------------------------|------------------------|-------------|-------------|---------------------------|-----------------------------|
| Doubly-robust variable selection in high dimensions | National Science Foundation | 09/1/2023 - 08/31/2026 | External    | PI          | \$46,154                  | \$28,846                    |

### *Past*

| <b>Name</b>                                                    | <b>Funding agency</b>       | <b>Period</b>         | <b>Type</b> | <b>Role</b> | <b>Annual direct cost</b> | <b>Annual indirect cost</b> |
|----------------------------------------------------------------|-----------------------------|-----------------------|-------------|-------------|---------------------------|-----------------------------|
| Testing and estimation for multi-modality single cell genomics | National Science Foundation | 08/01/2021-07/31/2024 | External    | PI          | \$37,948                  | \$23,718                    |
| Statistical Software for Single Cell CRISPR Screens            | Analytics at Wharton        | 2021-2023             | Internal    | PI          | \$25,000                  | \$0                         |